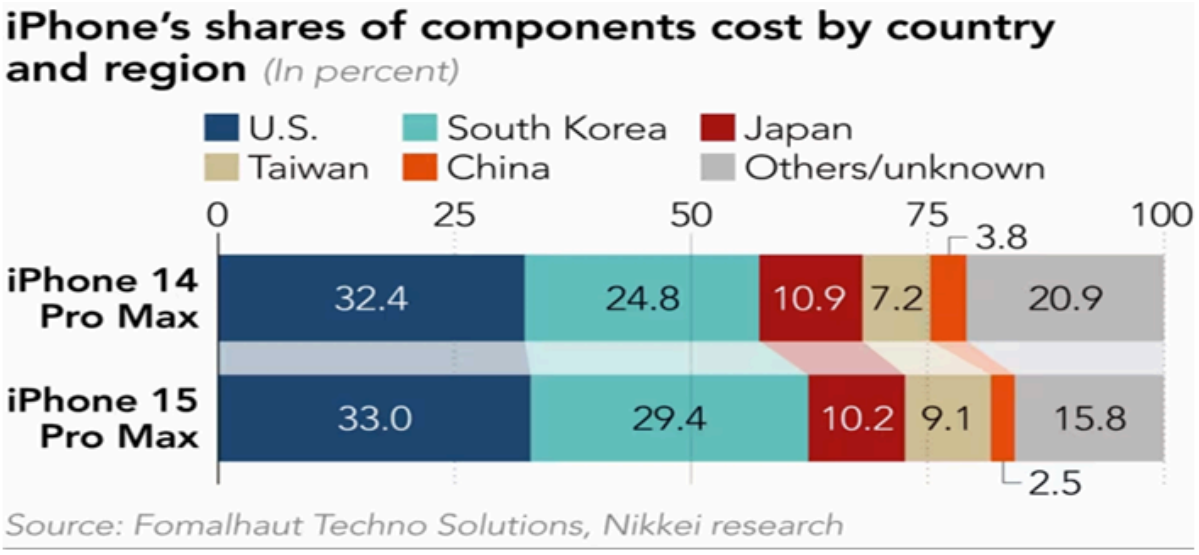
Blanquerna–Ramon Llull University, School of Communication and International Relations, Official Master's degree in Advanced Studies in International Affairs

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The following figure shows the list of component providers for the IPhone:



Consigne : Answer the following questions using analytical argumentation:

1. Identify and explain the economic phenomenon illustrated by the graph.
2. Discuss the main factors and drivers triggering this systemic process.
3. Explain the emerging International Labour Division resulting from this process.

When consumers purchase a smartphone, they often associate it with a single country of origin. If it is an Iphone, it is an American Product, if it's Samsung, it's Korean.Yet, behind all the process, lies a more complex chain, from conception to production, all integrated in the global economy. The Iphone, labelled as *"Design in California"* is in reality, a fragmented process of production, burst in multiple continents. Our document is a graph extracted from a Japanese Electronics Consultancy (Fomalhaut Techno Solutions) collaborating with Nikkei marketing research and consultancy firm(Nikkei Group), a stock market Japanese company. It demonstrates the distribution of iPhone component costs by country and region, makes this hidden geography of production visible and raises a fundamental question: how is value created and distributed in the contemporary global economy?

Our main thesis, defended in this essay, will be that the graph, exemplified the rise of Supply Chain Trade, meaning the entire system, involved in the production of a good or service, in this case, the Iphone, within the framework of Global Value Chain's (GVC) and that, this, is a defining part of 21st Century Capitalism. We are not in a neutral process of Globalization and efficiency, but, in a structural part of the economy, driven by technological change, global accumulation and national economic policies. Following on Baldwin and Lopez-Gonzalez's analysis on SCT (2014) and professor Lapavitsas and the EReNSEP (2023), established in 2016 and critical of contemporary capitalism, the essay will try to demonstrate that these dynamics are producing a new shift in International Division of Labor characterized by specialization, power relations and inequalities between global economies.

I. Identification and Explanation of the Economic Phenomenon Illustrated by the Graph

The Graph illustrates the form of production of the Iphone, traditionally, the epitome of Global Capitalism, so widespread it is in the world, and the form of production of GVC, in which the different stages need cooperation and coordination across borders. This phenomenon, reflects the growing dominance of production networks, that are, both specialized in the production of material, and at the same time fragmented. If we follow the graph more precisely, we can see the rise of supply-Chain Trade (SCT), defined by Richard Baldwin and Javier Lopez-Gonzalez, both specialized in International Economics, as an international trade dominated by exchanges of intermediate goods rather than final products (Baldwin and López-González, 2013). We clearly see that components of the Iphone cross borders multiple times, before assembly in Asia. The Display is made in South Korea, the Flash memory in Japan, the Chip is from design in California and produced in Taiwan, as well as the assembly, in Zhengzhou, China and a small part in India (represented in the Graph by the gray part "Unknown"). This confirms that contemporary globalisation is structured around transnational production networks rather than nationally integrated industries.

Following the reading of the Graph, we can see three central characteristics of Contemporary Globalisation in the production. First, the functional specialization within the Value Chain. The United States captures the largest share of total production costs, margin between 32,4 for the Iphone 14 Pro Max to 33% in the 15 pro max model, of total material production, reflecting its dominance in high value-added activities such as research and development, software, product design and marketing. Concerning Advanced manufacturing, such as the innovative technologies East Asian Economies are in pole position. South Korea specialises in semiconductors and display technologies, Japan in precision components and materials, and Taiwan in chip fabrication. China, despite its central role in assembly, captures a relatively limited share of total value. Asian Economies Production represent, globally roughly 50% of the global Iphone production. But, if we concentrate on the pathway, we can see a continuity between Iphone 14 pro max production and Iphone 15 pro Max where figures are

globally the same. Academically, this supports Baldwin and López-González's (2013) argument that supply-chain trade creates deeply rooted production structures that are costly and difficult to reorganise. The last, but not least factor of our first part argument, is the unequal value capture (=process of recovering).
While production is geographically dispersed, the most profitable segments of the value chain remain concentrated in economies that control intangible assets and coordination functions. This demonstrates what Milberg and Winkler showed, meaning that participation in GVCs does not automatically translate into proportional economic gains (Milberg and Winkler, 2013). The 20.9 % regarding the Iphone 14 pro max, or the 15.8% in the iphone 15 pro max, translate in global economy terms, that the gain of this part of the global value chain, is not equally distributed, as in the roughly 30% of the United States, or the 25-30% of South Korean production and return on production gains .

Within a conventional economic framework, supply-chain commerce illustrates the benefits of efficiency resulting from comparative advantages related to specific tasks. By breaking down production processes, companies are able to reduce expenses, enhance productivity, and provide lower-priced goods to customers (Grossman and Rossi-Hansberg, 2008; Baldwin and López-González, 2013).Nonetheless, critical political economy perspectives contest this view. Lapavitsas and the EReNSEP Writing Collective (2023) contend that modern capitalism is defined by increased hierarchies, monopolistic dominance, and inequitable value extraction. According to this perspective, global value chains are not simply tools of efficiency; rather, they are institutional frameworks that perpetuate imbalanced power dynamics between central and peripheral economies. The value chain of the iPhone reinforces this essential understanding, as nations engaged in the most sophisticated technological or labor-intensive activities extract much lower value compared to those overseeing design, finance, and intellectual property.

II. Main Factors and Drivers Triggering This Systemic Process

The imbalanced allocation of value displayed in the chart results from a systemic mechanism influenced by technological, economic, institutional, and geopolitical elements. Even though we see various countries in the supply chain production of the Iphone, the graph shows that 60 per cent of the overall worth is accounted for by the collective contributions of the United States and South Korea, whereas China's portion stays under ten percent. This imbalance highlights fundamental factors rooted in the current framework of global capitalism. But why ? What are the main factors that are triggering this systemic process ? How do we measure value, within the Global Value Chain ?
Technological competencies serve as a fundamental factor influencing value appropriation within Global Value Chains. If a country is the only one, determining the capability of producing a particular product within GVC,it is crucial to the good development of products. In the case of the production of the Iphone, the case of nano-technologies, and the core role of Taiwan, or, South Korea'n case, that retains around 28–30 percent of the overall component expenditure associated with the iPhone 15 Pro Max, particularly in semi-conductors and technologies. This is one example of the 'specialization' process, that is a result of global policies, that establishes, research ecosystem, high investment particularly in R&D, and strong industrial coordination. The core example would be Taiwan, which shares up to 9.1% cost of the production. In contrast, China's contribution, amounting to roughly 5–7 per cent, is largely concentrated in assembly and low-margin manufacturing processes. This again, echoes Baldwin and Lopez-Gonzalez argument, the assertion that trade within supply chains enables nations possessing technological benefits to focus on high-value activities while delegating labor-intensive processes to others (2013). In this sense, countries that would not have, or obtain

technological concentration, in the case of the iphone, nano-technologies, would act as a barrier to entry, and this would generate concentration of value, rather than the diffusion of the GVC.

In addition to technological factors, we can see that the graph indicates changes occurring in modern capitalistic dynamics of GVC. The US accounts for 33 to 35 percent of the overall component expenses, but a small share of production is made within the borders. This shows the non-physical assets, such as the brand recognition, goodwill, or the data of the production of the Iphone 14 pro Max and 15 pro Max, or even the Software, and branding. According to Greek economist, and professor at SOAS University of London, Costas Lapavitsas, with EReNSEP Writing Collective, in 2023, this arrangement is identified as a type of capitalism marked by centralized oversight of production networks devoid of direct ownership of manufacturing resources. The model of Apple, is the pinnacle of the theory. The management of the global production, the establishment of benchmarks, appropriation of profits, while offloading labor expanses and production uncertainty upon suppliers (Case of the model of the Apple Care for the production of Iphone). Consequently, economies that focus on manufacturing and assembly receive only a small portion of the overall value. Institutional and political elements are crucial in determining the GVC within production networks. Industrial policies, implemented in Asia, from the beginning of the 21th century, creates the significant portions acquired by South-Korea and Japan. They are focused on enhancing domestic technological competencies and advancing companies in key sectors. In contrast, China's restricted ability to capture value, notwithstanding its pivotal role in assembly, can be attributed to both institutional limitations and its standing within global intellectual property frameworks predominantly influenced by companies based in developed nations. According to Gereffi, Humphrey, and Sturgeon (2005), the governance frameworks within Global Value Chains influence the distribution of value and the allocation of decision-making authority. Therefore, the distribution illustrated in the graph results from not only market dynamics but also from institutional frameworks grounded in political contexts (free trade or not for example)

The last, but not the least, factor influencing this systemic process, that appears in a more subtle way, is the geopolitical tensions. Even though, when we read the graph, we see a slight change, between Iphone 14 pro max and Iphone 15 pro max, we see a re-orientation of supply chain slowly shifting from China to South-Korea, in response to geopolitical tensions and technological competition. Nevertheless, China’s ongoing participation, which contributes around 5 to 7 percent of the overall value, demonstrates the durability of international production systems. Baldwin and López-González (2013) highlight that trade within supply chains fosters profound interconnections that are expensive and challenging to break apart. Instead of indicating a decline in globalization, the data suggests a targeted restructuring limited by prevailing structural interdependencies. We can see the emergence of a new global system structured around functional specialization, and power relations.

III. The Emerging International Division of Labour in High Tech Capitalism resulting from this graph

The allocation of value throughout the supply chain of the iPhone exemplifies a significant shift in the international division of labor under modern capitalism. Instead of being predominantly organized by national industries or specific sectors, the international division of labor is progressively centered around dispersed productive activities integrated within global production networks. Gradually, a more fragmented Division of Labour is observed. Activities of great significance, including product development, system design, and management of intellectual property, are predominantly found within a small selection of developed nations, with the United States being the

most prominent example. In contrast, manufacturing and assembly processes are spread out across various regions in East and Southeast Asia. We are in a more functionally differentiated international division of labor, rather than the industrial Division of Labor, that was observed over the last two decades. The limited number of economies concentrate in the US, while manufacturing is a proper Asian characteristic. In this sense, and according to Durand, countries are integrated into global markets according to their position in the value-creation hierarchy rather than their overall level of industrialization (Durand, 2016). Therefore, activities of great significance, including product development, system design, and management of intellectual property, are predominantly found within a small selection of developed nations, with the United States being the most prominent example. In contrast, manufacturing and assembly processes are spread out across various regions in East and Southeast Asia

However, one critique that could be made to this new IDL, could be the fact that it is inherently hierarchical. As Kaplinski explained, dominance over the most lucrative areas of manufacturing is held by companies and nations that can monopolize expertise, benchmarks, and organizational efforts. (Kaplinski, 2005).
In this sense, the significant proportion of value associated with components in the United States illustrates its preeminence in profit-generating endeavors shielded from direct competition in labor. In contrast, intermediate economies, in the case of our graph, South Korea or Japan, specialize in technologically advanced elements, however, are still limited in their structure due to their reliance on leading companies for entry into markets and for setting prices. In this sense, the power relations will still be unbalanced, between production and supply depending on the value of each product of the GVC. If we take for example, the case of China, that represents between 2.5 and 3.8% of component cost of production depending on the Iphone 14 Pro Max or Iphone 15 Pro Max. China, remains in a lesser role, than the other countries, in the GVC, a role in which they concentrated on assembly processes and manufacturing with low profit margins. Its small proportion of component value verifies that mere involvement in industry does not ensure the retention of value, supporting Susan Strange's (1988) assertion that influence in the global economy is wielded through structural dominance rather than the quantity of production. China has a very strategic role in the GVC and plays a key structural role in ensuring a good manufactured product and control.

Following this logic of GVC and Structural Specialization, within the Capitalistic system, that we can extrapolate from our document, we see a contradiction within our own system. As Milberg and Winkler explained, on the one hand, involvement in international production networks may promote the transfer of technology, generate job opportunities, and encourage industrial education. But, on the other hand, the division of production functions often confines nations to structurally unequal positions, restricting their potential for advancement within the value chain.(Milberg & Winkler, 2013). Working conditions illustrate these imbalances. Employment that offers high wages and stability is primarily found in design and research and development facilities, while unstable, low-paying jobs prevail in assembly areas. This geographical division allows companies to exploit differences in labor costs and regulatory environments, diminishing the effectiveness of collective bargaining on a worldwide scale. This will be one negative facet of the growing interdependence within the International Division of Labour (Harvey, 2005). Consequently, we could argue that the International Division of Labor perpetuates global disparities while masking them beneath themes of innovation, efficiency, and advancements in technology. But one thing is certain, we are in a time where the International Division of Labour finds itself a central focus of our societies.

However, although recent geopolitical conflicts, disruptions in supply chains, and industrial policy efforts indicate a potential restructuring of the International Division of Labor (IDL), the findings suggest more of a reorganization rather than a fundamental change. Production processes may undergo geographical relocations, and certain manufacturing capacities could be brought back to domestic shores or diversified, yet dominance over the high-value sectors continues to be heavily centralized.

As noted by Arrighi (1994) and Izquierdo Brichs (2017), a comprehensive transformation requires not only a shift in the location of production but also a realignment of authority concerning value generation and appropriation. Absent this realignment, the global division of labor is expected to remain largely unchanged: adaptable in terms of geography, yet fixed within its hierarchical structure.

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